

Simple, highly reliable network communication  
system



## ISE2005D Unmanaged Industrial Ethernet Switch

· Plug-and-play

· DIN-rail

· Fanless

· IP30

# 1. Product Overview

The ISE2005D is designed for power, transportation, industrial control, and demanding industrial environments.

## Product Features:

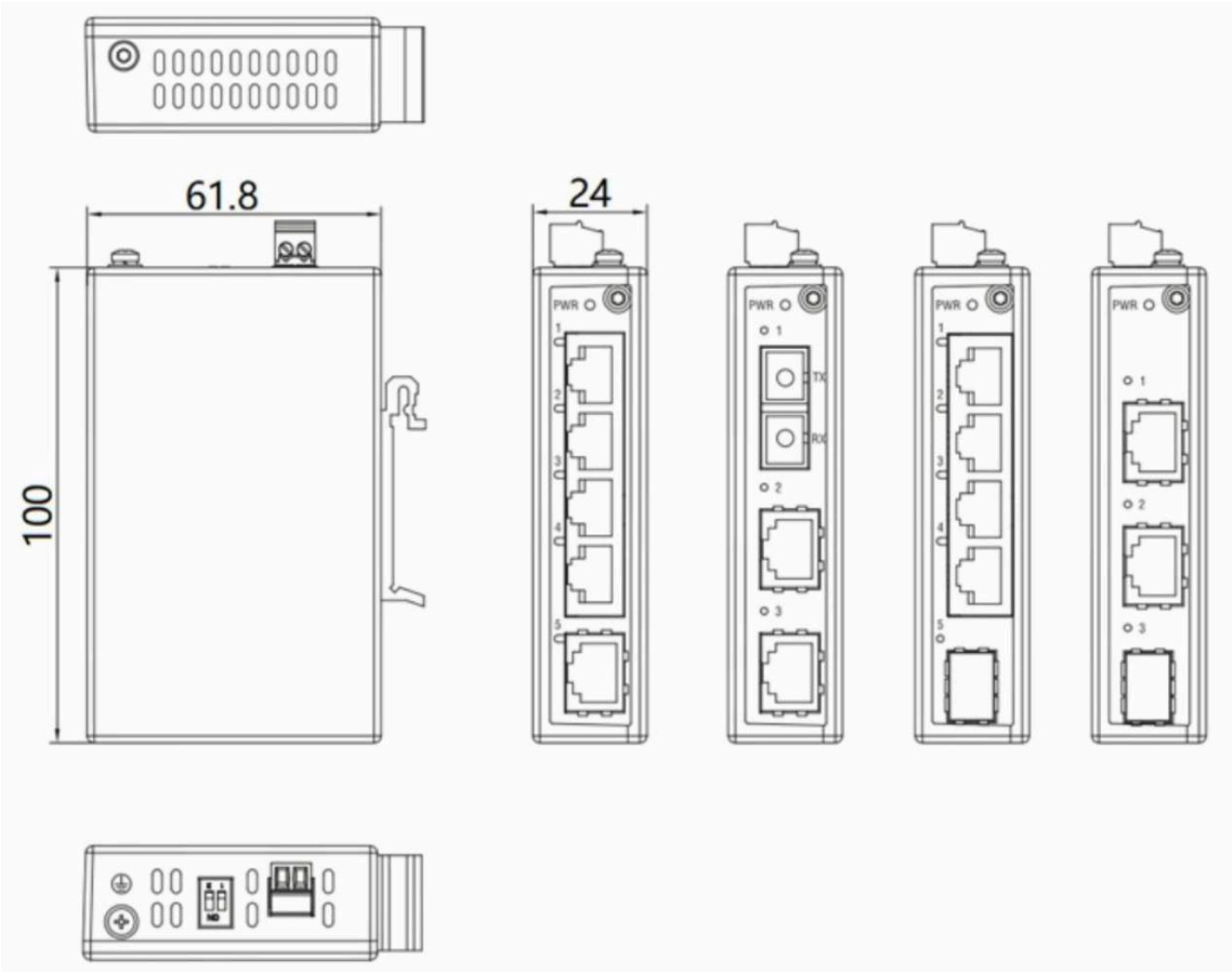
- **Robust Design:** Fully enclosed seamless metal casing, fanless design, IP30 protection rating.
- **Wide Adaptability:** Wide-temperature (-40 ~ +75°C) and wide-voltage (9.6-60VDC & 18-30VAC) design.
- **International Standard:** Complies with FCC, CE, ROHS, and UL standards with excellent EMC.
- **Smart Switching:** QoS and Broadcast Storm Protection (BSP), configurable via DIP switch.
- **Easy Deployment:** Plug-and-play, DIN rail mountable, compact size for quick deployment.

## Key Technical Specifications

| Parameter             | Specification                                                  |
|-----------------------|----------------------------------------------------------------|
| Type                  | Layer 2 Unmanaged Industrial Ethernet Switch                   |
| Ports                 | 5 x 10/100BaseT                                                |
| Switching Performance | 1.6 Gbps backplane, Store-and-Forward, 1K MAC                  |
| Dimensions / Weight   | 24 × 100 × 61.8 mm (0.94 × 3.94 × 2.43 in) / 0.23 kg (0.51 lb) |
| Power                 | 9.6-60 VDC & 18-30 VAC, 5 W                                    |
| Environment           | -40 °C ~ +75 °C (-40 °F ~ +167 °F) operating, IP30             |
| EMC                   | IEC 61000-4-2/3/4/5/6/18 Class 3, IEC 61000-4-8 Class 5        |

| Parameter      | Specification |
|----------------|---------------|
| Certifications | CE, FCC, UL   |

## 2. Product Dimensions



ISE2005D

- Note:
- 1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
  - 2. All dimensions are approximate and **for reference only**.
  - 3. The dimensions shown in the figure **shall not be used for production or processing**.
  - 4. Dimensions must comply with part and manufacturing tolerance requirements.
  - 5. Dimensions are subject to change without notice.

### 3. Hardware Specifications

| Category/Parameter          | Specification                           |
|-----------------------------|-----------------------------------------|
| <b>Physical Performance</b> |                                         |
| Enclosure                   | Fully enclosed seamless metal enclosure |
| Dimensions (W × D × H)      | 24 mm × 100 mm × 61.8 mm                |
| Weight                      | 0.23 kg (0.51 lb)                       |
| Mounting Method             | DIN-rail mounting                       |
| Cooling Method              | Fanless cooling                         |
| Ingress Protection          | IP30                                    |
| Storage Temperature         | -40 °C ~ +85 °C (-40 °F ~ +185 °F)      |
| Operating Temperature       | -40 °C ~ +75 °C (-40 °F ~ +167 °F)      |
| Humidity                    | 5 ~ 95% (non-condensing)                |
| <b>Hardware Performance</b> |                                         |
| Backplane Bandwidth         | 1.6 Gbps                                |
| Processing Type             | Store-and-Forward                       |
| MAC Table Size              | 1K                                      |
| Packet Buffer Size          | 448 Kbit                                |
| Switching Delay             | <10 μs                                  |

| Category/Parameter                     | Specification                                                                                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DIP Switch                             | Quality of Service (QoS), Broadcast Storm Protection (BSP)                                                                                                                                         |
| <b>Power Parameters</b>                |                                                                                                                                                                                                    |
| Operating Voltage                      | 9.6–60 VDC & 18–30 VAC                                                                                                                                                                             |
| Power Consumption                      | 5 W                                                                                                                                                                                                |
| Overcurrent Protection                 | Supported                                                                                                                                                                                          |
| Reverse Polarity Protection            | Supported                                                                                                                                                                                          |
| <b>Electromagnetic Characteristics</b> |                                                                                                                                                                                                    |
| EMS                                    | IEC(EN)61000-4-2, Class 3; IEC(EN)61000-4-3, Class 3<br>IEC(EN)61000-4-4, Class 3; IEC(EN)61000-4-5, Class 3<br>IEC(EN)61000-4-6, Class 3; IEC(EN)61000-4-8, Class 5<br>IEC(EN)61000-4-18, Class 3 |
| <b>Certifications</b>                  |                                                                                                                                                                                                    |
| Certifications                         | CE, FCC, UL                                                                                                                                                                                        |
| <b>Quality Assurance</b>               |                                                                                                                                                                                                    |
| Warranty Period                        | 5 years                                                                                                                                                                                            |
| MTBF                                   | 35 years                                                                                                                                                                                           |

## DIP Switch Settings

Users can use the DIP switch on the device panel to enable/disable Quality of Service (QoS) and Broadcast Storm Protection (BSP) functions. Fast Ethernet ports support 10/100 Mbps auto-negotiation.

| Category/Parameter                     | Specification                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality of Service (QoS) – On          | Enables the QoS function, processes packet priorities in 2 WRR queues. Priority 7,6,5,4 → Queue 1 (WRR Weight 16); Priority 3,2,1,0 → Queue 0 (WRR Weight 1) |
| Quality of Service (QoS) – Off         | Disables the QoS function                                                                                                                                    |
| Broadcast Storm Protection (BSP) – On  | Enables broadcast storm protection for all ports (allows max 200 broadcast packets per second)                                                               |
| Broadcast Storm Protection (BSP) – Off | Disables the broadcast storm protection function                                                                                                             |

## 4. Ordering Guide

| Model            | Description                                                                                                                                                                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISE2005D-P-5T-24 | 5-port Layer 2 unmanaged Industrial Ethernet Switch. 5 × 10/100 BaseT Ports. IP30 Protection Class, Operating Temperature from -40 °C to +75 °C (-40 °F to +167 °F). Single power input: 12/24/48 VDC (9.6~60VDC) or 24 VAC (18~30 VAC) (Only connect to class 2 power supply). |

## 5. Contact Us

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